

OBSERVATIONAL ESTIMATORS IN SME EVALUATION: TESTING MODERN NONEXPERIMENTAL METHODS

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Randomized controlled trials are prohibitively expensive in developing countries, forcing researchers to rely on nonexperimental data. This paper tests whether modern observational methods can recover experimental treatment effects for small businesses, with limited sample and unobserved firm differences. Using data from Sebrae na Sua Empresa in Rio de Janeiro and an observational control group, I compare 21 nonexperimental approaches against an experimental benchmark (ITT = -0.1426) using 4,051 firms linked across survey and employment records.

Results show observational methods can work when researchers preserve treatment structure carefully. Double/Debiased Machine Learning with separate treatment arms in competitive markets achieved near-perfect replication ($\beta = -0.147$). Combining matching with panel methods also performed well ($\beta = -0.152$). Standard cross-sectional matching methods showed persistent upward bias ($\beta = +0.026$ to $+0.356$). Overall, observational analysis works in business evaluation when it accounts for unobserved firm and intervention differences.

INTRODUCTION

Sebrae na Sua Empresa is a program that evaluates small businesses in Rio de Janeiro. Between 2017 and 2019, the World Bank randomly assigned 4,051 firms to three groups: one received a diagnostic report plus financial literacy training, another received a report plus peer comparisons, and the third received only the report. The study found no measurable effect on business practice adoption or training participation (ITT = -0.1426).

Although the experiment showed no effect, the data it generated is valuable. The research team linked firm surveys to Brazil's official employment registry (RAIS), creating a dataset where the true treatment effect is known. This is an ideal setting to test nonexperimental methods in developing countries, given the prohibitive costs of running randomized trials. Prior work on this dataset tested classical methods like propensity score matching. This paper expands that analysis to include 21 different approaches, from traditional matching to modern machine learning, to determine which methods researchers can trust.

LITERATURE REVIEW

A foundational study by LaLonde (1986) compared nonexperimental estimators to an experimental benchmark from a job training program. The nonexperimental methods failed badly and cast doubt on their efficacy, producing estimates far from the

true effect. Heckman, Ichimura, and Todd (1998) diagnosed why: the treatment and control groups differed on unobserved characteristics, measurement tools were inconsistent, and selection into treatment was not random on observables. Recently, Imbens and Xu (2025) revisited this question and found overlap on observed characteristics and covariate balance enabled nonexperimental methods to replicate experimental results. This suggests LaLonde's failure was not inevitable but reflected insufficient attention to design diagnostics.

Modern statistical methods provide different approaches to address selection bias in high-dimensional data. Double/Debiased Machine Learning uses a technique called orthogonalization to adjust for many confounders without introducing regularization bias. Entropy Balancing reweights the control group to match the treatment group's characteristics. Coarsened Exact Matching groups units on rounded characteristics and matches within groups. Difference-in-Differences uses panel data to remove permanent firm differences.

Testing these methods in small business evaluation is especially important. McKenzie and Woodruff (2014) document that business training programs typically show modest effects against noisy outcomes, and unobserved skills may influence the likelihood of participating and adoption of new practices. Benchmarking nonexperimental methods in this challenging setting directly tests whether they work under realistic constraints.

DATA & METHODS

I linked 4,051 firms across three data sources: the World Bank survey (baseline characteristics and 2019 outcomes), Brazil's employment registry RAIS (firm data), and Sebrae program records (treatment assignment). The main outcome is a score measuring adoption of 24 business practices (ranging 0–24). I used 8 core covariates—firm age, employee count, firm type, local market density, owner education, owner business experience, owner gender, whether owner or manager responded, and owner risk aversion—plus missing value indicators for a total of 24 factors.

Though the treatment group is the same as the experiment, the control group is different. I selected firms unrelated to the experiment by matching with a larger set of firms. Given that observational methods wouldn't have access to randomized experimental data, this helps replicate the existing limitations.

I evaluated 21 different estimators organized into three groups. The first group applies matching and reweighting using data from the RAIS registry as a nonexperimental control group. This includes simple unadjusted comparisons, propensity score matching using several algorithms, coarsened exact matching, entropy balancing, and doubly robust methods. The second group uses the panel structure of RAIS (data from 2015–2019) to estimate difference-in-differences, which

removes permanent unobserved firm characteristics. The third group applies Double Machine Learning, a modern machine learning method that estimates treatment effects while avoiding bias from regularization.

RESULTS

Figure 1 shows all 21 estimates plotted against the experimental benchmark. Three clear patterns emerge: certain methods worked reliably, others showed the right direction but wrong magnitude, and a large group failed completely.

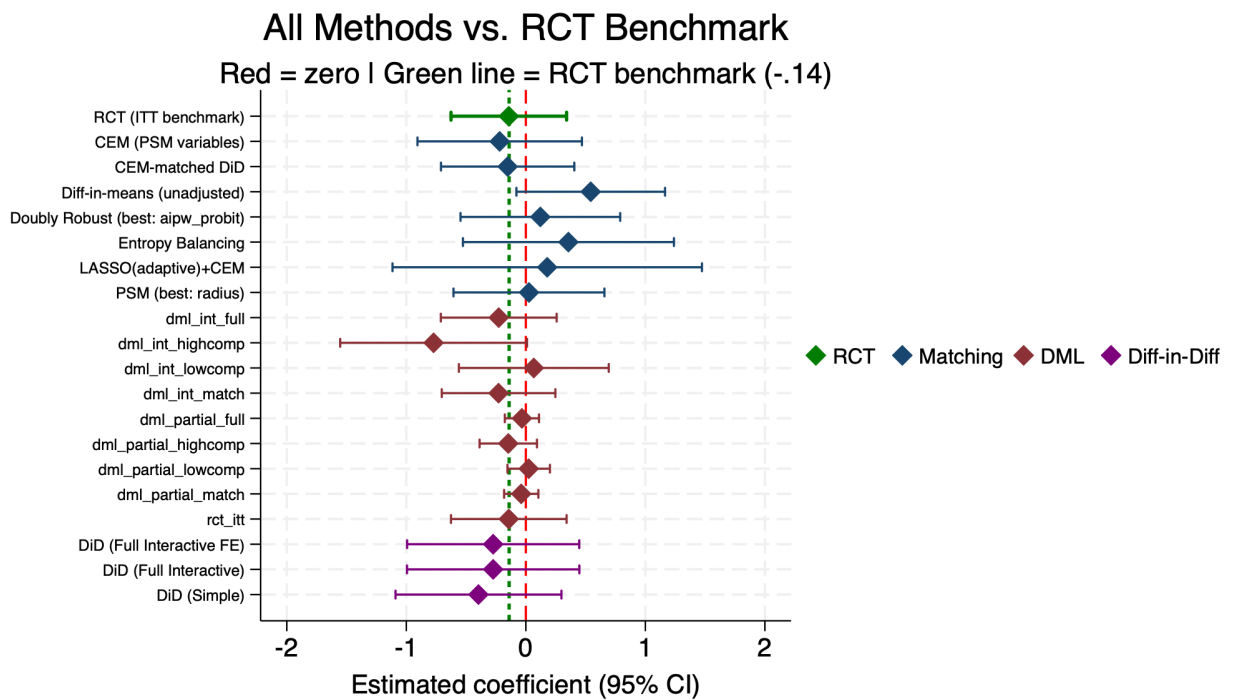


Figure 1

Two approaches achieved accurate estimates. Double Machine Learning applied to separate treatment arms in high-competition markets nearly perfectly replicated the benchmark. The key to its success was preserving the distinction between the two different interventions. When machine learning was forced to treat all programs as a single binary treatment instead, performance collapsed across all sample types, with estimates becoming wildly unstable and sometimes reversing direction entirely. This dramatic failure suggests machine learning needs accurate information about how different programs actually differ.

The second successful approach combined coarsened exact matching with difference-in-differences. By matching firms on a small set of key covariates and then comparing outcomes before and after the program within matched groups, this hybrid method achieved near-benchmark accuracy. The success came from only selecting the most important confounders and implementing a panel structure to eliminate permanent firm differences.

All difference-in-differences specifications correctly identified that the program had a negative effect, but they overstated how negative. Even when covariate interactions were included to account for different time trends, the estimates remained overly negative. Comparing firms before and after the program does eliminate some unobserved bias, but something else remains uncontrolled—possibly unexpected economic shocks that affected firms differently over time.

Every standard cross-sectional matching method reported false positive effects, despite the true effect being zero or negative. Propensity score matching, entropy balancing, and doubly robust approaches all pushed estimates into positive territory. Simple unadjusted comparison was worst, but all conventional approaches showed the same directional bias. This consistent pattern demonstrates a fundamental problem: reweighting control units to match treatment group characteristics on observed data cannot overcome unobserved owner traits—ambition, talent, work ethic—that both influence who chooses to participate and whether they adopt practices regardless of the program.

CONCLUSION

This paper tests a practical question: can researchers in developing countries rely on nonexperimental methods when randomized trials are too expensive? Yes, but only with careful methodological choices.

The most reliable approach combined three elements. First, preserve the distinction between different interventions instead of treating all programs as a single binary treatment. Machine learning can learn how different programs affect adoption differently, but only if not forced to collapse that information. Second, use panel data when available. Comparing each firm's outcomes before and after the program removes permanent differences between participating and nonparticipating firms.

Third, focus on key confounders rather than trying to use all available data. Exact matching on three important characteristics (owner type, baseline performance, formal employment) was generally better than reweighting all eight covariates.

Unfortunately, standard cross-sectional matching methods consistently reported false positive effects. Propensity score matching, entropy balancing, and other reweighting approaches were also biased by unobserved characteristics, demonstrating how reweighting on observed traits alone cannot address hidden bias.

This project has many limitations. Business practice scores are self-reported, introducing measurement noise that widens confidence intervals across all methods. Preserving separate treatment definitions in machine learning reduces the sample size for each estimate. The new control group was also reused for all analysis, rather than trying multiple subsets of unrelated firms. Future research should test whether these findings hold for other business programs and explore whether some firms benefited while others were harmed.

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